Highlights

The Boussinesq–compressible gap in steady two-dimensional buoyancy-driven enclosure convection is linear to leading order in the relative temperature contrast

- Boussinesq–compressible gap follows $R = A\varepsilon^m$ with $m = 1.02$ in velocity; θ is weak and flow-dependent, and the pressure exponent is fixed-gauge
- Relative temperature contrast is the only break-symmetry axis; a 24$\times$ Ma change is below solver resolution
- 627 paired solves calibrate the law; public data reproduce its amplitude at one contrast
- At $\varepsilon = 1.2$ integral Nu shifts 0.54% (median 0.58%) but local wall flux redistributes 4.25%
- Factored prior $D = \varepsilon P(X)$: 1.6$\times$ over plain and 1.87$\times$ channel-averaged over ε-blind networks